

Zachary Deutsch

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Education

DUKE UNIVERSITY

B.S.E in Mechanical Engineering. Minor in Machine Learning. Minor in Philosophy. GPA: 3.72
Relevant Coursework: Mechanics of Materials; Engineering Design; Statistics & Data Analysis;
Data Structures & Algorithms; Technical Communication; Physics I & II; Calculus I, II, & III; Linear Algebra

Durham, NC
Expected May 2027

UNIVERSITY OF OXFORD

Study Abroad Coursework in AI and Ethics

Oxford, England
June – August 2025

Experience

DUKE UNIVERSITY, HICKEY LAB

Machine Learning Engineer

- Build unsupervised ML pipelines for multi-organ cell type annotations of spatial-omics data with Dr. John Hickey
- Undergraduate Research Assistantship grant recipient

Durham, NC
November 2024 - Present

TEL AVIV UNIVERSITY, DATA MODELING & ANALYSIS LAB

Machine Learning Research Intern

- Evaluated quantile-based feature extraction method for sEMG time series classification with Dr. Neta Rabin

Tel Aviv, Israel
May – August 2025

DUKE UNIVERSITY, DEPARTMENT OF RADIOLOGY

Mechanical Engineering Intern

- Led 16 undergraduates to develop catheter placement training system with Dr. Jonathan Martin
- 1st place out of 100 teams at Fortin Foundation Bass Connections Showcase

Durham, NC
September 2024 – May 2025

SHIRLEY RYAN ABILITY LAB, CENTER FOR BIONIC MEDICINE

Data Scientist

- Analyzed data from \$1.5M DoD-funded clinical trial on VR treatment for phantom limb pain with Dr. Levi Hargrove

Chicago, IL
June – August 2024

Activities

DUKE UNIVERSITY, DEPARTMENT OF BIOMEDICAL ENGINEERING

Biomedical Engineering Society Treasurer

- Raised \$15k and led 14 undergraduates at BMES 2025 Annual Meeting

Durham, NC
September 2024 - Present

Conferences

- Poster Presenter, “sEMG Hand and Wrist Movement Time-Series Classification: Evaluating QUANT and Cosine Features”, BMES Annual Meeting, 2025
- Co-author, “Computational Framework for Multi-organ Cell Type Annotation of HuBMAP Single-cell Spatial-omics Data”, BMES Annual Meeting, 2025
- Co-author, “Computational Framework for Multi-organ Cell Type Annotation of HuBMAP Single-cell Spatial-omics Data”, HuBMAP Annual Meeting, 2025
- Oral Presenter, “Virtual Reality for Health Education Advanced Learning”, Fortin Foundation Bass Connections Showcase, 2025
- Poster Presenter, “Computational Pipeline for Cell Type Annotations of HuBMAP Spatial-omics Data”, Duke Undergraduate Research Symposium, 2025
- Poster Presenter, “Liquid Level Detection for bCPAP Respirators in Low-Resource Settings”, Duke Engineering Design Expo, 2024

Skills, Awards, & Interests

Technical: AI/ML, Python, Java, Microcontrollers, Circuitry, 3D Printing, CAD

Languages: Conversational Proficiency in Japanese and Spanish

Awards: National Youth Science Academy, 1st place J.P. Morgan GenerationTech, Illinois Science & Technology Coalition Scholarship

Interests: Espresso, Liszt, Stoicism, Nietzsche, Cycling